Two-Paper Package — Introduction (v0.2, RE-SCOPED per K1714/K1715). From one forced geometry: the color structure and the correct one-loop running of the strong force. Paper B proves the substrate D_IV^5 is the unique bounded symmetric domain meeting the physics criteria (Cartan elimination, criteria-innocent), forcing $dim\_C = 5$ and $N\_c = 3$ — so the color NUMBER 3, its SO(3), and baryon number are forced; the SU(3) gauge DYNAMICS is imported, not derived (#108/K1677) — we do NOT say 'the gauge group is derived.' Paper A, on that forced substrate, derives the asymptotic-freedom SIGN blind from the a_2 heat-kernel coefficient (an $\mathbb{R}^4$-valid dynamical credential) and exhibits a parameter-free compactification gap $2(d\_S-1) = 6$ on the boundary sphere ($d\_S = n\_C-1 = 4$). HONEST SCOPE, STATED UP FRONT: the compactification gap is a free-Hodge / Kaluza–Klein gap ($\propto 1/a^2$, $\to 0$ as $a \to \infty$), NOT the interacting Yang–Mills (Clay) mass gap (fixed in the $\mathbb{R}^4$ limit) — the two scale oppositely, and the Clay mass gap is LEFT OPEN. We never claim to have solved a Millennium Problem. v0.2 draft.

Lyra (Claude Opus 4.8) — Casey Koons, PI; Grace, Elie, Cal, Keeper

2026-08-19 Wednesday (date-verified)

One Geometry: a Forced Color Number and a Blind One-Loop Running

Introduction to the two-paper package (re-scoped)

Paper A — The Strong Sector from D_IV^5: a Forced Gauge Group, Blind One-Loop Running, and a Parameter-Free Compactification Gap (not the Yang–Mills mass gap). Paper B — D_IV^5 Substrate Uniqueness (Cartan Elimination).

The claim, in one sentence

From a single geometry that is *forced* rather than chosen, the strong force's color structure — the number 3, its SO(3), its baryon number — *and* the direction of its running coupling follow, with no adjustable dimensionless numbers and one dimensionful ruler that every physical theory must pay.

We are explicit about what is *not* in that sentence: the SU(3) gauge dynamics (imported, not derived — geometry forces the color *number*, not the Yang–Mills dynamics; #108/K1677) and the Yang–Mills (Clay) mass gap (we do not derive it and do not claim it).

The bright-high-schooler version

Two things about the strong nuclear force are usually fed *into* the theory: *why* it's built on the symmetry SU(3) ("three colors"), and the fact that it gets *weaker* at very short distances (called asymptotic freedom — the discovery that won a Nobel Prize). This package gets both *out* of one shape.

- Paper B shows the shape isn't a choice. Among all the "nicely curved bounded spaces" you could build a physics on, exactly one meets the minimal requirements — and that one, D_IV^5, has the number 3 baked in. So "three colors" isn't put in by hand; it's what the only workable shape has.
- Paper A shows that on that same shape, the force *automatically* comes out getting weaker at short distances — the sign of the effect falls out of the ge-

ometry with nothing put in. That's a real dynamical result, and it holds on ordinary flat space, not just on the shape.

Now the honest part, and we put it first, not last. The shape *also* has a "gap" — a smallest vibration, the way a drum of finite size has a lowest note. That gap comes out to a clean 6, set purely by the boundary being a 4-dimensional sphere. It's real and it's parameter-free — but it is not the famous "mass gap" of the Clay Millennium Problem, and it's important to say why: our gap is just the drum being *finite* (make the drum infinitely large and the note drops to zero), whereas the real Yang–Mills mass gap *stays fixed* no matter how large you make space. They behave in opposite ways as you grow the space — so ours cannot be the Clay object, and we don't pretend it is. The Clay mass gap remains the hard, open problem it has always been.

The logical spine (formal)

Paper B — the substrate is forced (and with it, the gauge group). A *criteria-innocent* uniqueness: among irreducible Hermitian symmetric domains, the requirements to carry physics single out D_IV^5 = SO$_0$(5,2)/[SO(5)×SO(2)], forcing dim_C = 5 and N_c = 3 (the dimension-free output of the scalar-Wallach equality N_c = rank^2−1 = 3, via T1829). So the color number 3, the real-type SO(3), and baryon number are forced — but the SU(3) gauge *dynamics* is imported, not derived (the gauge principle / inner fluctuation; #108/K1677). This answers "why exactly 3 colors" (forced); it does not claim to derive the Yang–Mills dynamics, and we do not say "the gauge group is derived."

Paper A — what the forced substrate delivers. 1. The one-loop running ($\mathbb{R}^4$-valid). The asymptotic-freedom sign $\beta_0 > 0$ lands blind from the a$_2$ Seeley–DeWitt coefficient of the fielded gauge operator — a short-distance statement, valid on flat $\mathbb{R}^4$, not a finite-volume artifact. This is the genuine strong-sector dynamical content. 2. A parameter-free compactification gap (a KK fact, not the mass gap). On the boundary (S^4×S^1)/$\mathbb{Z}_2$, the *free* Hodge Laplacian on transverse 1-forms has lowest eigenvalue 2(d_S − 1) = 6 (in 1/a^2 units), dimension-rooted (U(1)/SU(3)/SU(5) all give 6 — the gauge group sets the degeneracy, not the eigenvalue). It is a genuine geometric number, but it is a Kaluza–Klein compactification gap, and it scales 6/a^2 → 0 as a → ∞ — opposite to the fixed Clay gap, hence not the Yang–Mills mass gap (K1714).

The only dimensionful input is the ruler (a Planck-scale unit — the one input GR pays as G and QM as $\hbar$, provably the theoretical floor). Dimensionless free parameters: zero.

Honest scope (stated up front, not buried)

- Derived / $\mathbb{R}^4$-valid: the color structure — number 3, SO(3), baryon# (Paper B); the one-loop asymptotic-freedom sign (Paper A) — no chosen dimensionless input. *(The SU(3) gauge dynamics is imported, not derived; #108/K1677.)*

- Parameter-free geometry, honestly labeled: the compactification gap $2(d\_S-1) = 6$ (a KK fact, $\propto 1/a^2$); the compact-boundary glueball ratios (suggestive, consistent with lattice; identification with physical glueballs open).
- Left open (the hard problem, K940): the Clay $\mathbb{R}^4$ Yang–Mills mass gap. Our compact-boundary gap is *not* a route to it — the two scale oppositely (a settled negative). A reconstruction route (OS/Rehren) remains a separate open candidate.
- Never claimed: "we derived the Yang–Mills mass gap"; "we solved a Millennium Problem." We say the opposite, on the first page.

Two standing guards the package holds

- The compactification gap is not the Clay mass gap. Free-Hodge/KK gap $\propto 1/a^2 \to 0$ in the $\mathbb{R}^4$ limit; the Clay gap is fixed. Opposite scaling; we state it ourselves.
- The number 6 is dimension-rooted — not "colors," not the scalar Casimir C_2. It is $2(d\_S-1)$, fixed by the boundary-sphere dimension; the gauge group sets only the degeneracy. Conflating it with the color number or with $C_2 = n\_C+1$ were errors, corrected in drafting.

Reading order

Read Paper B first (why the arena and the gauge group are forced), then Paper A (the running and the KK gap on that arena, and why the Clay gap is left open). Paper A's Section 0 states the KK-vs-Clay scaling up front.

Package intro v0.2 (RE-SCOPED, K1714/K1715). Paper A = v1.1 (strong-sector re-scope: color number + SO(3) + baryon# forced, SU(3) dynamics imported; AF sign derived blind; boundary gap = KK, not Clay). Paper B = v0.6 (substrate + N_c=3 forced, Path A / T1829). One geometry, zero chosen dimensionless inputs, one ruler; SU(3) dynamics imported; the Clay mass gap left open. Ships on Keeper-PASS + Cal-vet + both voices + Casey GO. Nothing pushed; CP existence-only.

— Lyra, Wed 2026-08-19 (date-verified).